

Clinical Investigation

Temporal Evolution and Dose-Volume Histogram Predictors of Visual Acuity After Proton Beam Radiation Therapy of Uveal Melanoma



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Summary

Proton beam radiation therapy (PBRT) of uveal melanoma (UM) provides effective local control with the potential for vision preservation. In this study, long-term functional follow-up of a large, uniformly treated and assessed UM cohort at specified time points after PBRT reveals preservation of excellent vision in 45% of patients and identifies 28-GyE exposure of the macula and optic nerve as independent DVH predictors of post-PBRT vision loss among those with initially favorable visual acuity.

Purpose: To perform an in-depth temporal analysis of visual acuity (VA) outcomes after proton beam radiation therapy (PBRT) in a large, uniformly treated cohort of uveal melanoma (UM) patients, to determine trends in VA evolution depending on pretreatment and temporally defined posttreatment VA measurements; and to investigate the relevance of specific patient, tumor and dose-volume parameters to posttreatment vision loss.

Methods and Materials: Uveal melanoma patients receiving PBRT were identified from a prospectively maintained database. Included patients ($n = 645$) received 56 GyE in 4 fractions, had pretreatment best corrected VA (BCVA) in the affected eye of count fingers (CF) or better, with posttreatment VA assessment at specified post-PBRT time point(s). Patients were grouped according to the pretreatment BCVA into favorable ($\geq 20/40$) or unfavorable ($20/50$ – $20/400$) and poor (CF) strata. Temporal analysis of BCVA changes was described, and univariate and forward stepwise multivariate logistic regression analyses were performed to identify predictors for VA loss.

Results: Median VA follow-up was 53 months (range, 3–213 months). At 60-month follow up, among evaluable treated eyes with favorable pretreatment BCVA, 45% retained BCVA $\geq 20/40$, whereas among evaluable treated eyes with initially unfavorable/poor BCVA, 21% had vision $\geq 20/100$. Among those with a favorable initial BCVA, attaining BCVA of $\geq 20/40$ at any posttreatment time point was associated with subsequent maintenance of excellent BCVA. Multivariate analysis identified volume of the macula receiving 28GyE ($P < .0001$) and optic nerve ($P = .0004$) as independent dose-volume

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